

Figure 2.3 Neutron decay according to the quark model (simplified by omitting the intermediate vector boson, which decays to an electron and an antineutrino).

experimental evidence, has been so great that its name, the Standard Model of matter, is justified.

The force between two particles arises from the exchange of a "mediator" that carries the force at a finite speed: one of the particles emits the mediator, the other absorbs it. The mediator propagates through space and, briefly, is not lodged with either particle. These mediators have the same properties as the elementary particles — mass, electric charge, spin — so physicists often call them particles as well, even though their role in nature is quite different from that of the elementary particles. The mediators are the mortar that binds the particle building blocks together.

Three kinds of mediators — or exchange forces, as we have called them — are known in the nuclear world: photons which are involved in the electromagnetic force, gluons which are the mediators of the strong nuclear force, and the weak force mediators, which underlies the radioactive decay. We know now a great deal about the photons and the gluons, but little about the weak force. The weak force mediators are the W^+, W^- and Z^0 "particles". A high energy electron is supposed to be able to emit a Z^0, and then a positron can absorb the electron: the particles annihilate each other, leaving the Z^0 momentarily free. Afterwards the Z^0 must decay back into a pair of elementary particles, such as an electron and positron, or a quark and an antiquark. Z^0 particles are produced in high energy proton-antiproton colliders; around 1983 researchers at the CERN and SLAC laboratories were able to determine the Z^0 mass to 91.2 GeV. This is the heaviest known unit of matter today.

2.7. EXERCISES

2.1. Which type of mesons are released in high energy particle interactions, and why?

2.2. (a) What kinds of forces exist in nature? (b) How does the weak interaction manifest its properties?

2.3. (a) What are bosons and how do they differ from fermions? (b) Does the difference have any practical consequence?

2.4. What proof exists that the photon has matter properties?

2.5. How can the neutrino be detected?

2.8. LITERATURE

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